MILITARY SPECIFICATION

STANDING WAVE RATIO-POWER METER ME-165( )/G

1. SCOPE

1.1 This specification covers one type of combined standing wave ratio and radio-frequency (rf) power output meter designated Standing Wave Ratio-Power Meter ME-165( )/G. (See 6.1 and 6.5.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571 —Solder: Lead Alloy, Tin Lead Alloy, and Tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.
QQ-S-781 —Steel Strapping, Flat.
UU-T-111 —Tape, Paper Gummed (Sealing and Securing).
PPP-B-601—Boxes, Wood, Cleated Plywood.
PPP-B-621—Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636—Boxes, Fiber.
PPP-F-320—Fiberboard; Sheet, Stock and Cut Shapes.
PPP-T-76 —Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing).

MILITARY

MIL-P-116 —Preservation, Methods of.
MIL-B-131 —Barrier Material, Water Vaporproof, Flexible.
MIL-V-173 —Varnish, Moisture-and-Fungus Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment.
MIL-T-713 —Twine and Tape, Lacing and Tying (For Use in Electrical and Electronic Equipment).
MIL-D-3464 —Desiccants, Activated, Bagged, Packaging use Static Dehumidification.
MIL-W-6858 —Welding: Aluminum, Magnesium, Non-Hardening Steels or Alloys, and Titanium, Spot, Seam and Stitch.
MIL-M-10304—Meters, Electrical Indicating Panel Type, Ruggedized.
MIL-M-10578—Metal Conditioner and Rust Remover (Phosphoric Acid Base).
MIL-M-13231—Marking of Electronic Items.
MIL-F-14072 —Finishes for Ground Signal Equipment.
MIL-S-19500 —Semiconductors, Diode.

FSC 6625

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STANDARDS

FEDERAL

595—Colors.

MILITARY

MIL-STD-22 —Welded-Joint Designs.
MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129—Marking for Shipment and Storage.
MIL-STD-169—Extreme-Temperature Cycle.
MIL-STD-170—Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-252—Wired Equipment, Classification of Visual and Mechanical Defects.
MS-20003 —Indicator, Humidity Card, Three Spot, Impregnated Areas.

DRAWINGS

SIGNAL CORPS

SC-A-46439 —List of Accessories for Package Tester.
SC-B-61578 —Grounding to Chassis.
SC-DL-849717—Drawing and Data List for Standing Wave Ratio-Power Meter ME-165( )/G.

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS

H29—Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C.)

3. REQUIREMENTS

3.1 Description. The equipment is used for measuring voltage standing wave ratios (vswr) between 1.5 and 20 megacycles (mc) in a 50-ohm coaxial feed system at a power level of 600 watts or less. This unit is also used for measuring power output in the same system. Usually, this equipment is connected with radio-frequency cable to transmitter and to the antenna feed line; an internal switch allows selection of the resistor bank for power measurement.

3.2 Preproduction samples. The contractor shall furnish two preproduction samples of the equipment for approval if required by the invitation for bids and contract. (See 4.3.)

3.3 Construction. The equipment shall be constructed in accordance with Drawing SC-DL-849717.

3.4 Parts, materials, and processes.

3.4.1 *Parts, materials, and processes; general.* Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements, as specified herein. Ratings and derated values of parts and materials shall not be exceeded when the equipment is subjected to specified service conditions (See 3.19).

3.5 Cleaning.

3.5.1 *Parts.* After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.5.2 *Units.* After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.6 Finish. The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4) The final paint film on Type I

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surfaces shall be semi-gloss gray, color 26152, in accordance with Federal Standard Number 595.

3.7 **Marking.**

3.7.1 *General.* Marking shall conform to Specification MIL-M-18231. (See 4.4.) Front panel marking shall be Group I as described in that specification.

3.7.2 *Visibility of markings.* Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.3 *Serial numbers.* The Standing Wave Ratio-Power Meter ME-165( )/G shall have serial numbers.

3.8 **Riveting.** Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.9 **Securing of parts.**

3.9.1 *General.* Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the equipment is subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.9.2 *Securing of parts by threaded fasteners.* Nylok or other screws with plastic devices, and Loctite or similar sealants, shall not be subjected to temperatures in excess of 250° Fahrenheit (F.) (as during baking of paint). Loctite or similar sealants shall not be used where the connection must be electrically conductive; where used, such materials shall be applied in accordance with the manufacturer's instruction.

3.9.3 *Brittle material.* Brittle castings, or parts made of ceramic or other brittle material, shall not be over-constrained by the means used to secure them. Not more than three points of contact shall be used between the brittle part and the mounting surface, unless the mating faces are accurately machined. Threaded holes in ceramic material shall not be employed for assembly or mounting of parts.

3.10 **Soldering**

3.10.1 *Solder.* Solder shall conform to Specification QQ-S-571, and the equipment drawings.

3.10.2 *Acid or acid salts.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.10.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

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3.11 Tropicalization of material. Material shall be treated as follows:

3.11.1 *Treating materials.* Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.11.2 *Toxicity.* Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.11.3 *Flexibility.* The treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.11.4 *Statement of treatment.* The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2.)

3.12 Welding. Wherever practicable, welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Stop, stitch, and seam welds shall conform to Specifications MIL-W-6858.

3.12.1 *Cleaning prior to welding.* Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.12.2 *Process.* Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.12.3 *Cleaning after welding.* Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-[illegible]. Acid used for cleaning shall be completely neutralized and removed.

3.13 Wiring and cabling. Wiring and cabling shall be neat and sturdy.

3.13.1 *Cabling.* Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance. The cabling of wires shall be effected by tape or twine conforming to Specification MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.13.2 *Slack.* Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected.

3.13.3 *Protection.* Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

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Wires shall not be bent sharply where they enter insulation material.

3.13.4 *Support.* Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner. (*Note:* When shielding on wire or cable is unprotected by an outer insulation, adequate support is necessary to prevent the shielding from coming in contact with exposed terminals or conductors.) Twine or tape shall not be used for securing wire and cable.

3.13.5 *Clearance.* Clearance between wires or cables, and parts such as resistors, shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearing between solder connections or bare conductors, on terminal boards or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.13.6 *Splicing.* Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.13.7 *Inductive and capacitive effects.* Wires and cables shall be so located that inductive and capacitive effects, unless used as a design feature, will be the minimum practicable.

3.13.8 *Connections.* Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL–V–173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.13.9 *Grounding.* Ground connection to shields and to other mechanical parts, except the chassis or frame, shall not be made to complete electrical circuits but only to eliminate high-potential alternating-current points.

3.14 **Grounding to chassis.** Grounding to chassis shall conform to Drawing SC–B–61578.

3.15 **Screws, other threaded devices, and related parts.**

3.15.1 *Form and fit.* Unless otherwise specified, screw threads shall conform to Handbook H28; the Unified Thread Form and Series shall be Class 2A or 2B fit, and the American National Thread Series shall be Class 2 fit.

3.15.2 *Thread engagement and extension.* The length of thread engagement for tapped holes in steel or hard brass shall at least equal the nominal thickness of corresponding machine-screw nuts in Sizes No. 12 and smaller and semifinished hexagon regular nuts in sizes ¼ inch and larger (see 3.15.1 above). Thread engagement for tapped holes in aluminum, magnesium, or other soft material shall be at least 50 percent greater than the thread engagement specified for steel or hard brass. Where permitted by design of the equipment, screws and bolts shall extend at least 1½ threads beyond the nut, but no more than 1½ threads plus ⅛ inch for screws up to 1 inch in length and 1½ threads plus ¼ inch for screws over 1 inch in length.

3.16 **Semiconductors (diode).** These electron devices (semiconductors) shall conform to Specification MIL–S–19500. (See 4.4.)

3.17 **Controls.** All movable controls shall operate properly without binding or other

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undue restriction. Controls shall not be assembled in a misalined condition.

3.18 **Meter.** The meter shall conform to Specification MIL–M–10304 and the equipment drawings. (See 4.4.)

3.19 **Service conditions (see 4.6).** The equipment shall meet the following service conditions; where a test is referenced, meeting the test shall be considered as compliance with the requirement.

3.19.1 *Operation.* Continuous use, for a period of one year, under the worldwide environmental conditions specified in 3.19.2 through 3.19.6, with no more than normal maintenance and replacement of parts. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.19.2 *Temperature* (see 4.6.1).

(a) *Operating:* Ambient temperature in the range of +150° F. to —40° F. (The 150° F. temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

(b) *Nonoperating:* Exposure in the range of +160° F. to —80° F. exposure of the high temperature extreme need not exceed 4 hours, and at the low temperature extreme need not exceed 24 hours, at any one time.

3.19.3 *Relative humidity* (see 4.6.2). Exposure to 97 percent relative humidity for 20 hours and 100 percent relative humidity, with condensation, for 4 hours for each 24 hour period over the period of years stated in 3.19.1. Meeting the tests of 4.6.2 shall be considered compliance with this requirement

3.19.4 *Elevation* (see 4.6.3).

(a) *Operating:* Up to 10,000 feet above sea level.

(b) *Nonoperating:* Up to 25,000 feet above sea level.

3.19.5 *Orientation.*

(a) *Operating* (see 4.6.4). Any orientation up to 20 degrees from normal operating position (that is, with front of chassis elevated, depressed, tilted left, and tilted right).

(b) *Nonoperating:* Storage in any position for a period of two years. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.19.6 *Vibration, bounce, and shock.* As described in 3.20.

3.20 **Vibration, internal.** The amplitude of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second. (See 4.6.5.)

3.20.1 *Bounce and shock.* The equipment shall be capable of meeting the requirements of table II.

TABLE II. *Bounce and shock*

Para heading	*Para. no.*	*Performance after test*
Bounce	4.6.6	Specified performance. No physical damage.
Shock; bench-handling	4.6 7	Specified performance (Note A) No physical damage.

Note A The equipment shall meet the following requirement 5.24 Operational inspection

3.21 **Bounce preconditioning.** The equipment shall be capable of meeting the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning of 4.7. (Also see 4.5.)

3.22 **Electrical requirements** (see 4.8).

3.22.1 *Power and frequency ranges* When tested as specified in 4.8.1, the equipment shall not be damaged because of heat dissipation of 550 watts at 1.5, 3, 10, and 20 mc.

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3.22.2 *Standing wave ratio.* When tested as specified 4.8.2, the standing wave ratio of the ME-165( )/G at the frequencies specified shall not read higher than the center of the Green area on the Meter scale.

3.22.3 *Indication accuracy* (see 4.8.3). The power meter readings noted during the test specified in 4.8.1 shall not be more than ± 10 percent from the actual power dissipated.

3.23 **Interchangeability.** Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.9.) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.24 **Operational inspection.** To determine that the equipment is operable, prior to packaging and at such other times as required, the operational test specified in 4.10 shall be performed. Meter readings shall be satisfactory.

3.25 **Technical literature and running spare parts.** Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2).

3.26 **Workmanship.** The equipment shall be manufactured and assembled in accordance with the equipment drawings and applicable portions of the following paragraphs herein:

3.4 Parts, materials, and processes; general.
3.5 Cleaning.
3.8 Riveting.
3.9 Securing of parts.
3.10 Soldering.
3.11 Tropicalization of material.
3.12 Welding.
3.13 Wiring and cabling.
3.15 Screws, other threaded devices, and related parts.
3.17 Controls.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Classification of inspection.** Inspection shall be classified as follows (see 6.3):

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)
(b) Inspection covered by subsidiary documents. (See 4.4)
(c) Quality conformance inspection of preparation for delivery. (See 4.12)

4.3 **Preproduction inspection.** This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table III; the inspection specified in the subsidiary documents covering the items listed in 4.4; and the inspection specified for group A, and group B (see tables IV, and V, respectively). The proproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, and (3) shock, benchhandling, other preproduction inspection may precede, follow, or be interspersed between the foregoing.

TABLE III. *Preproduction inspection*

Inspection	Requirement paragraph	Inspection paragraph
Temperature	3.19.2	4.6.1
Relative humidity	3.19.3	4.6.2
Elevation	3.19.4	4.6.3
Orientation	3.19.5	4.6.4
Vibration, internal	3.20	4.6.5
Bounce	3.20.1	4.6.6
Shock, bench-handling	3.20.1	4.6.7

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4.4 Acceptance inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary document as part of the inspection before preparation for delivery:

Item	Where required
Finish	3.6
Marking	3.7
Semiconductor	3.16
Meter	3.18

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.2.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government—at its discretion—may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7.) Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, or group B inspection, except preproduction samples shall be preconditioned after final assembly. (See 3.21.)

4.5.1 *Group A inspection.* This inspection, including sampling, shall conform to table IV and the ordinary inspection procedures of Standard MIL–STD–105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection (4.10) shall be last (see 6.5).

TABLE IV. *Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and mechanical	3.26	4.11	2.5 dphu*	10.0 dphu
Electrical				
Power and frequency range	3.22.1	4.8.1	2.5 dphu	**
Standing wave ratio	3.22.2	4.8.2		
Indicator accuracy	3.22.3	4.8.3		
***Operational inspection	3.23	4.10	10%	**

* Defects per hundred units.
** Defects in these characteristics are of major category
*** This test need not be performed, as part of Group A inspection, if units are packaged immediately after electrical tests

4.5.2 *Group B inspection.* This inspection, including sampling, shall conform to table V and to the special procedures for small-sample inspection of Standard MIL–STD–105. The reduced inspection procedure shall be R–1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection.

TABLE V. *Group B inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL (See 4.5.2.1)
Interchangeability	3.23	4.9	6.5%

4.5.2.1 *Group B sampling plan.* The acceptable quality level (AQL) shall be 6.5 percent defective, as specified in table V, and the inspection level shall be L–7 for normal and L–5 for reduced inspection

4.5.2.2 *Order of inspection within group B.* Group B inspection shall be performed in any order which is satisfactory to the Government.

4.6 Service conditions tests (see 3.19).

4.6.1 *Temperature test* (see 3.19.2). The equipment shall be subjected to the temperature cycle shown on Standard MIL–STD–169. Measurements shall be taken at step

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10. The equipment shall meet the operational inspection test of 4.10 at the given temperature step.

4.6.2 *Moisture-resistance test* (see 3.19.3). The equipment shall be tested as follows:

4.6.2.1 *Test conditions.*

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Perform the entire test with the chassis removed from its enclosure.

4.6.2.2 *Test method.*

(a) Dry at 130° ± 5° F. for 24 hours.

(b) Condition at 77° ± 5° F. and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.6.2.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.6.2.3 shall be made during the periods shown on the standard and shall comply with 4.6.2.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ± 5° F. and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.6.2.3.

4.6.2.3 *Performance.* There shall be no mechanical deterioration, as evidenced by corrosion of parts or binding of rotating or sliding parts. The following measurements shall be made:

Test para	*Measurement*
4.10	Operational inspection

4.6.2.4 *Failure.* If the equipment fails to meet the performance specified in 4.6.2.3 during cycling, it shall be readjusted once. If the equipment then fails to meet specified performance, or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, it does not pass the test.

4.6.3 *Elevation test* (see 3.19.4). The equipment, placed in its normal operating position in an altitude chamber, shall be tested as follows: (*Note:* To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

Paragraph	*Measurement*
4.10	Operational inspection

(b) Reduce the barometric pressure to 20.6 ± 0.1 inches of mercury (simulated 10,000 feet above sea level) and, after temperature and pressure stabilization for 2 hours, repeat the measurements specified in 4.6.3(a). The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 11.0 ± 0.1 inches of mercury (simulated 25,000 feet above sea level). Maintain this condition for 2 hours.

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(d) Return the chamber to 29.9 ± 0.1 inches of mercury. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.6.3(a). The equipment shall meet specified performance for these measurements.

4.6.4 *Orientation test* (see 3.19.5). The equipment shall be inclined for a minimum of 5 minutes in each plane (that is, with front of chassis elevated, depressed, tilted left, and tilted right) to an angle of 20 + 3, —1 degrees. During inclination in each plane, the equipment shall meet specified performance for the following measurements:

Paragraph	*Measurement*
4.10	Operational inspection

4.6.5 *Test for internal vibration.* Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.20.

(a) Secure the equipment directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.6.6 *Bounce test* (see 3.20.1). The equipment shall be tested on the package tester, Type 1000–SC, as made by the L.A.B. Corporation, Skaneateles, N.Y., or equal. Accessories shall be selected from those listed on Drawing SC–A–46439. The test shall be as follows:

(a) Secure the equipment to the 48- by 48-inch vehicular adapter plate, in the same manner as it would be mounted in a vehicle. The adapter plate shall be free of cracks and broken welds, and all unused holes except the 2-inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

(b) Cover the tester bed with a panel of ½-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface at 6-inch intervals around all four edges and at 8-inch intervals in a 6-inch square in the center.

(c) Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

(d) Operate the package tester, shafts in phase, for a total of 3 hours at 285 ± 1 revolutions per minute. Turn the adapter plate 90 degrees at the end of each 45 minutes, each time in the same direction.

4.6.7 *Shock test; bench-handling* (see 3.20.1). The equipment shall be prepared as for servicing, and placed in a suitable posi-

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tion for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat (a), with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.7 **Bounce preconditioning (see 3.21).** The unit shall be placed in its normal operation position on the table of the Type 1000–SC Package Tester as made by the L.A.B. Corporation, Skaneateles, N.Y., or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.8 **Electrical tests (see 3.22).**

4.8.1 *Power and frequency range test.* Connect the output of a 550-watt, 1.5 to 20 mc transmitter through Radio Frequency Cable CG–55( )/U, or equal, to the input of the ME–165( )/G. With the ME–165( )/G switched to the internal resistive load and with 550 watts of input power, note meter reading.

4.8.2 *Standing wave ratio test.* Using a 50-ohm non-inductive resistance load across the output of the ME–165( )/G, the vswr reading shall meet the requirement of 3.22.2 at 1.5, 3.0, 10 and 20 mc.

4.8.3 *Indication accuracy test.* Meter readings noted during the test performed in 4.8.1 shall be checked for accuracy using a Hewlett-Packard VTVM Model 410B, or equal. The requirement of 3.22.3 shall be met.

4.9 **Inspection for interchangeability.** The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.23. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies when such dimensions affect mating of parts.
(b) Dimensions of cavities, when such dimensions affect insertions of items.
(c) Location of fasteners on separable parts or assemblies which must mate, such as cases and mountings.
(d) Location of connectors, fasteners, slides, and mountings which receive mating parts; and location of the mating part.
(e) Size and form of special threads, if any.

4.10 **Operational test.** A minimum of 350 watts at 10 mc shall be applied to the ME–165( )/G. The meter reading shall be noted on the power measurement position of the equipment. A noninductive load shall be connected to the ME–165( )/G and the SWR meter reading noted. The requirements of 3.24 shall be met.

4.11 **Visual and mechanical inspection.** Equipment shall be examined for the defects listed in Standard MIL–STD–252.

4.12 **Quality conformance inspection of preparation for delivery.** Preparation for delivery shall be in accordance with Specification MIL–P–116 to determine conformance to the requirements of section 5.

4.13 **Rough handling test (Preparation for delivery).** When a rough handling test in

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accordance with Specification MIL-P-116 is required by the contract (see 6.2. e(3)), the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling:

Operational test 4.10.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Each Standing Wave Ratio Power Meter ME-165( )/G shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in Specification MIL-P-116 and described in the following paragraphs.

5.1.1.1 *Standing Wave Ratio Power Meter ME-165( )/G.* Each Standing Wave Ratio Power Meter ME-165( )/G shall be packaged Method 11b as follows: Cushion each Power Meter on all surfaces with cells or pads or both, fabricated of fiberboard type CF, class Weather resistant, Variety SW, conforming to Specification PPP-F-320. Place and secure the required amount of fully activated desiccant, conforming to Specification MIL-D-3464, in accordance with Formula 1 of Specification MIL-P-116, within the cushioning cells. Place the cushioned item within a close-fitting fiberboard box, type I, class 1, conforming to Specification PPP-B-636. Secure the box closure with gummed paper tape conforming to Specification UU-T-111. Place a humidity indicator card, conforming to Military Standard MS-20003, on top of the sealed box and secure in position with pressure-sensitive tape conforming to Specification PPP-T-76. Blunt all corners of the box. Any exterior stapled or stitched joints shall be covered with gummed paper tape conforming to Specification UU-T-111. Place the boxed unit within a water vaporproof barrier bag, class 1, conforming to Specification MIL-B-131. The barrier bag shall be of sufficient size to allow for three (3) additional seals. Heat seal the closure leaving a small opening for exhausting air. Exhaust the entrapped air and complete the seal. The time lapse between positioning of the desiccant and the completion of the barrier bag seal must be held to an absolute minimum. Place the water-vaporproofed item within a close-fitting fiberboard box, style RSC, type I, class 2, W5c, conforming to Specification PPP-B-636. Any stapling or stitching on the inside of the exterior box shall be covered with gummed paper tape, conforming to Specification UU-T-111, to prevent damage to the barrier. Closure shall be in accordance with the appendix of the box specification.

5.1.2 *Level C.* Each Standing Wave Ratio Power Meter ME-165( )/G shall be preserved and packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.1.3 *Package performance.* Package testing shall be performed in accordance with the requirements of Specification MIL-P-116. The rough handling and functional test will be required only when invoked in the bid request or contract.

5.2 Packing.

5.2.1 *Level A.* Place a quantity of Standing Wave Ratio Power Meter ME-165( )/G within a close-fitting nailed wood box, style 4, class 2, or class 2, type III, style A, cleated plywood box conforming to Specification PPP-B-621 or PPP-B-601 respectively. Fabricate the box to fit contents snugly. Closure shall be in accordance with the appendix of the box specification. Gross weight limitation of the shipping container shall not exceed 200 pounds.

5.2.1.1 *Metal strapping.* Shipping containers shall be strapped with flat strapping conforming to Specification QQ-S-781, type I class b, in accordance with the requirements of the appendix of the applicable box specification.

5.2.2 *Level B.* Standing Wave Ratio Power Meter ME-165( )/G shall be packed as specified in 5.2.1, except the nailed wood box

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shall be class 1 and the sheeted plywood box shall be type IV, class 1.

5.2.3 *Level C.* Standing Wave Ratio Power Meter ME–165( )/G shall be packed in shipping containers of the type, size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Rules or Regulations of the common carrier applicable to the mode of transportation.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use. Standing Wave Ratio Power Meter ME–165( )/G is intended to switch a 5-section whip antenna to a 50-ohm, 500-watt transmitter output. The ME–165( )/G will measure the standing wave ratio and power output in a 50-ohm coaxial feed line system. Usually, Radio Transmitter T–368( )/GR or BC–610–( ) is used, with the output of the ME–165( ) connected to an antenna tuner such as the BC–939–( ), CU–45( )/MRQ–2, or TN–339( )/GR—which in turn connects to a whip or long-wire antenna.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification, and any amendment thereto.
(b) Type required.
(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)
(d) The specific paragraphs of section 5 which are applicable to the particular procurement.
(e) For preproduction inspection, preproduction pack(s) as follows:
  (1) Makeup of pack(s).
  (2) Number of each kind of pack to be submitted.
  (3) Inspection that will be performed thereon—including rough handling test.
(f) Marking and shipping of samples.
(g) Place of final inspection.
(h) Technical literature required. (See 3.25.)
(i) Quantity of running spare parts required. (See 3.25.)
(j) Submission of the statement of treatment referenced in 3.11.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Definitions.

6.3.1 *Equipment divisions.* The terms used herein for equipment divisions conform to Standard MIL–STD–280.

6.3.2 *Definitions of inspection terms.* The inspection terms used herein conform to Standard MIL–STD–109. The following are abbreviated definitions selected from those given therein for immediate reference:

6.3.2.1 *Inspection.* Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.3.2.2 *Examination.* Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.3.2.3 *Testing.* Testing consists of determinations of compliance, using technical means.

6.4 Location of operational inspection. It is desirable that the operational inspection (4.19) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.5 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a

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letter identifying the particular design; for example: ME-165W/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract (see 1.1).

6.6 **Verification inspection.** Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 51-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION		CITY AND STATE
CONTRACT NO	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Typed or printed name and activity)	DATE

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